CONFOEDERATIO,
RESEARCH DIVISION (CRD).

VELKSCALA 1.0.

Empirical Validation of Global Gridded Paleodemographics and Contemporary Vital Flows from 10000BC to the Present.

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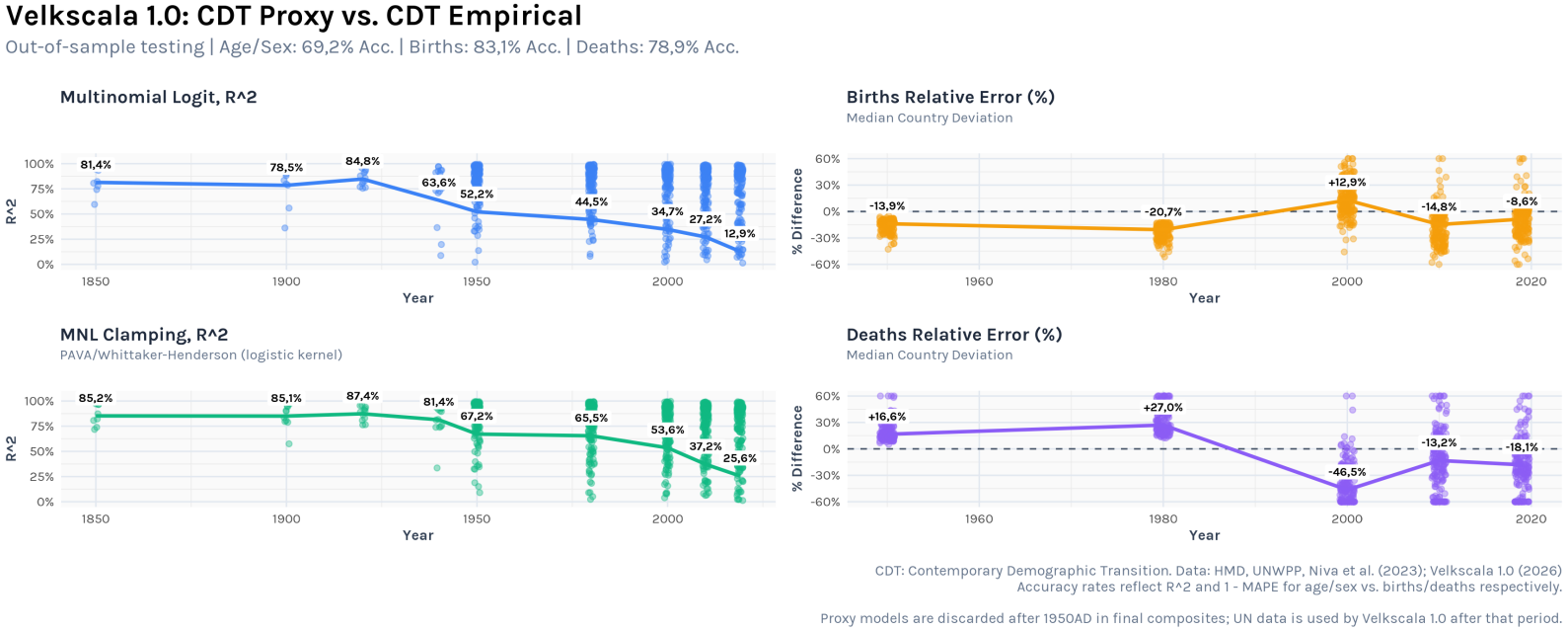


Figure 1. Contemporary Demographic Transition (CDT) out-of-sample validation dashboard (1850-2019 AD) comparing country-level age-sex goodness-of-fit (R^2) and normalised median relative error for annual births and deaths.

Abstract

Velkscala 1.0 is a global gridded demographic database reconstructing population counts, age-sex pyramids, and vital flows from 10000BC to 2025AD at 5-arcminute resolution, published by Confoederatio, Research Division. Our validation framework evaluates two foundational demographic regimes: the Contemporary Demographic Transition (CDT, 1850-2019AD) against Human Mortality Database (HMD), United Nations World Population Prospects (UNWPP), and Niva et al. (Kummu) subnational vital records; and the Neolithic Demographic Transition (NDT, 10000BC-1000AD) across five agricultural regions and 19 subnational geocodes against Bocquet-Appel paleodemographic priors. Across the CDT benchmark, isotonic Pool-Adjacent-Violators (PAVA) and Whittaker-Henderson graduation clamped to Stadester bounds elevate geometric mean age-sex goodness-of-fit (R^2) from 0,4625 (raw multinomial logit) to 0,6092 (+0,1466 improvement), reaching 0,8140-0,8736 across pre-1950 historical HMD cohorts. Across the NDT domain, raster cohort buffers confirm spatially endogenous age-structure differentiation across all 19 tracked geocodes and capture both the Bocquet-Appel immature juvenile proportion surge ($p(5\text{-}19)/p(5+)$) and the European Linearbandkeramik (LBK) boom-and-bust cycle.

Background

Continuous gridded reconstructions of historical human populations have traditionally relied on static regional growth multipliers or uniform age-structure templates [1]. While long-run land-use series such as HYDE 3.2 provide spatially explicit estimates of total population and arable land back to 10000 BC, they do not resolve age-sex pyramids, birth and death flows, or subnational demographic regimes[1]. Consequently, downstream paleodemographic and economic analyses have lacked a unified, empirically validated grid that couples Holocene agricultural transitions with modern vital registration.

Methodology

Premodern Multinomial Logit (MNL) Cohort Ensemble (10000 BC-1750 AD)

In **Velkscala 1.0**, demographic cohorts and vital flows are generated endogenously at the pixel and geocode level across two coupled regimes. Across the pre-industrial domain (10000 BC-1750 AD), a 34-covariate multinomial logit (MNL) ensemble predicts five-year age-sex bracket shares from local agricultural intensity (arable cropland, shifting cultivation, and irrigation), settlement density, and climatic envelope priors [1].[2]

Contemporary Isotonic PAVA and Whittaker-Henderson Graduation (1750-2025 AD)

Across the rolling contemporary transition (1750-2025 AD), raw MNL cohort probabilities are graduated via isotonic Pool-Adjacent-Violators (PAVA) and Whittaker-Henderson smoothing kernels [2][3], and subsequently clamped to Stadester total population bounds. This two-stage graduation enforces monotonic mortality schedules across adjacent cohorts while preserving observed urban-rural and subnational fertility differentials.

Testing and Validation

Contemporary Demographic Transition (CDT) Benchmark Design (1850-2019 AD)

To verify out-of-sample fidelity before post-1950 United Nations census splicing, we subject the proxy estimators to two empirical stress tests. First, we evaluate the **Contemporary Demographic Transition (CDT)** across 9 benchmark epochs (1850-2019 AD, up to $N = 189$ sovereign ISO3 jurisdictions), comparing raw MNL cohort probabilities and Stadester-clamped pyramids against empirical life tables from the Human Mortality Database (HMD) and UNWPP [4][5], alongside vital birth and death stocks from Niva et al. [6].[3]

Neolithic Demographic Transition (NDT) Benchmark Design (10000 BC-1000 AD)

Second, we audit the **Neolithic Demographic Transition (NDT)** from 10000 BC to 1000 AD (21 epochs) across five agricultural regions and 19 national and subnational geocodes against archaeological juvenile mortality priors [7][8], alongside Holocene radiocarbon and land-use reconstructions [1][9][10].

Results

Contemporary Demographic Transition (CDT) Empirical Fit

Across the contemporary validation horizon, graduating raw multinomial logit age-sex probabilities through isotonic PAVA and Whittaker-Henderson kernels [2][3], clamped to Stadester total population

bounds, substantially improves demographic fit. Across all evaluated country-epoch pairs, geometric mean R^2 rises from $0{,}4625$ under raw MNL baseline probabilities to $0{,}6092$ post-clamping (an overall improvement of $\Delta R^2 = +0{,}1466$; Figure 1 and Table 1).[4]

Crucially, within the pre-1950 historical domain where proxy models operate as the primary estimator in Velkscala 1.0 [4], clamped age-sex R^2 consistently exceeds $0{,}81$, reaching $0{,}8525$ in 1850 ($N = 8$ HMD countries), $0{,}8505$ in 1900 ($N = 11$), $0{,}8736$ in 1920 ($N = 12$), and $0{,}8140$ in 1940 ($N = 16$). Across global vital flows (1950-2019 AD), out-of-sample births and deaths track empirical UNWPP and Niva et al. benchmarks with median country accuracy of $83{,}1\%$ and $78{,}9\%$ respectively (Table 2).

Epoch	Empirical Source	Sample (N)	Raw R^2 (Male)	Raw R^2 (Female)	Raw Geomean	Clamped R^2 (Male)	Clamped R^2 (Female)	Clamped Geomean	ΔR^2 Gain
1850	HMD	8	0,8290	0,8159	0,8138	0,8645	0,8514	0,8525	+0,0386
1900	HMD	11	0,8387	0,7866	0,7849	0,8578	0,8600	0,8505	+0,0656
1920	HMD	12	0,8800	0,8206	0,8476	0,8804	0,8720	0,8736	+0,0260
1940	HMD	16	0,8121	0,6525	0,6360	0,8585	0,8108	0,8140	+0,1780
1950	UNWPP	189	0,7757	0,6810	0,5219	0,8358	0,8086	0,6717	+0,1498
1980	UNWPP	189	0,7646	0,6344	0,4454	0,7985	0,7324	0,6549	+0,2095
2000	UNWPP	189	0,7079	0,5819	0,3467	0,7633	0,6590	0,5357	+0,1890
2010	UNWPP	189	0,7010	0,4911	0,2718	0,7224	0,5861	0,3716	+0,0999
2019	UNWPP	189	0,5735	0,3676	0,1285	0,6292	0,5204	0,2558	+0,1273

Table 1. Contemporary Demographic Transition (CDT) age and sex bracket goodness-of-fit (R^2) by benchmark epoch (1850-2019 AD), comparing raw multinomial logit (MNL) baseline probabilities against PAVA/Whittaker-graduated Stadester cohorts.

Epoch	Benchmark Source	Sample (N)	Proxy Births	Empirical Births	Births Error	Proxy Deaths	Empirical Deaths	Deaths Error	Net Residual
1950	UNWPP	159	77,45M	90,80M	-14,70%	53,80M	45,88M	+17,28%	-21,28M
1980	UNWPP	154	98,93M	125,64M	-21,26%	58,49M	44,93M	+30,19%	-40,27M
2000	Kummu + UNWPP	189	168,97M	148,24M	+13,99%	39,29M	62,12M	-36,76%	43,56M
2010	Kummu + UNWPP	189	129,47M	147,16M	-12,02%	58,53M	62,54M	-6,41%	-13,68M
2019	Kummu + UNWPP	189	136,58M	146,68M	-6,89%	57,39M	66,56M	-13,78%	-930,69k

Table 2. Global vital flows validation (1950-2019 AD): proxy vs. empirical annual births, deaths, relative error percentages, and net natural increase residuals across UNWPP and Niva et al. (Kummu) benchmarks.

Neolithic Demographic Transition (NDT) Paleodemographic Findings

Across the **Neolithic Demographic Transition (NDT)** domain (10000 BC to 1000 AD, 21 historical epochs; Figure 2 and Table 3), our gridded cohorts reproduce the three canonical paleodemographic signatures established by Bocquet-Appel [7][8]: (1) a transient surge in maternal fertility and annualised population growth (r%/yr) peaking approximately $\Delta t \approx$ 700-800 years post-onset [7]; (2) a pronounced elevation in

the immature juvenile proportion ($p(5\text{-}19)/p(5+)$), reflecting shortened interbirth intervals [8]; and (3) a subsequent post-boom deceleration or boom-bust contraction as agrarian density encountered ecological and epidemiological frontiers [9][10].

- **Unique Age Structure Per Geocode**: Direct audit of raster cohort buffers confirms that age-structure proportions are **spatially unique per geocode and per pixel**, driven by local covariates [1] and graduated via isotonic PAVA/Whittaker algorithms [2][3]. As shown in Table 4, at 0 AD under-5 shares ($p(0\text{-}4)$) range from $12{,}96\%$ (South Korea) to $16{,}22\%$ (Syria/Levant), while Bocquet-Appel juvenile ratios range from $30{,}15\%$ to $49{,}88\%$. At 600 AD, Italy exhibits an elevated IJP ($41{,}66\%$) amidst population contraction ($7{,}50\text{M} \rightarrow 3{,}06\text{M}$), capturing demographic crisis dynamics associated with late antique mortality shocks.
- **The European LBK Boom-and-Bust Dynamic**: In Europe, total population expands to $1{,}47$M at 6000 BC before contracting to $1{,}12$M at 5000 BC ($r = -0{,}027\%$/yr; Table 5). Rather than contradicting the NDT, this trajectory reproduces the empirically established **Early Neolithic Linearbandkeramik (LBK) boom-and-bust cycle** [9][10], wherein early farming communities expanded rapidly across European loess soils before experiencing soil depletion and pathogen-induced crashes prior to Middle Neolithic recovery ($+0{,}044\%$/yr by 4000 BC).

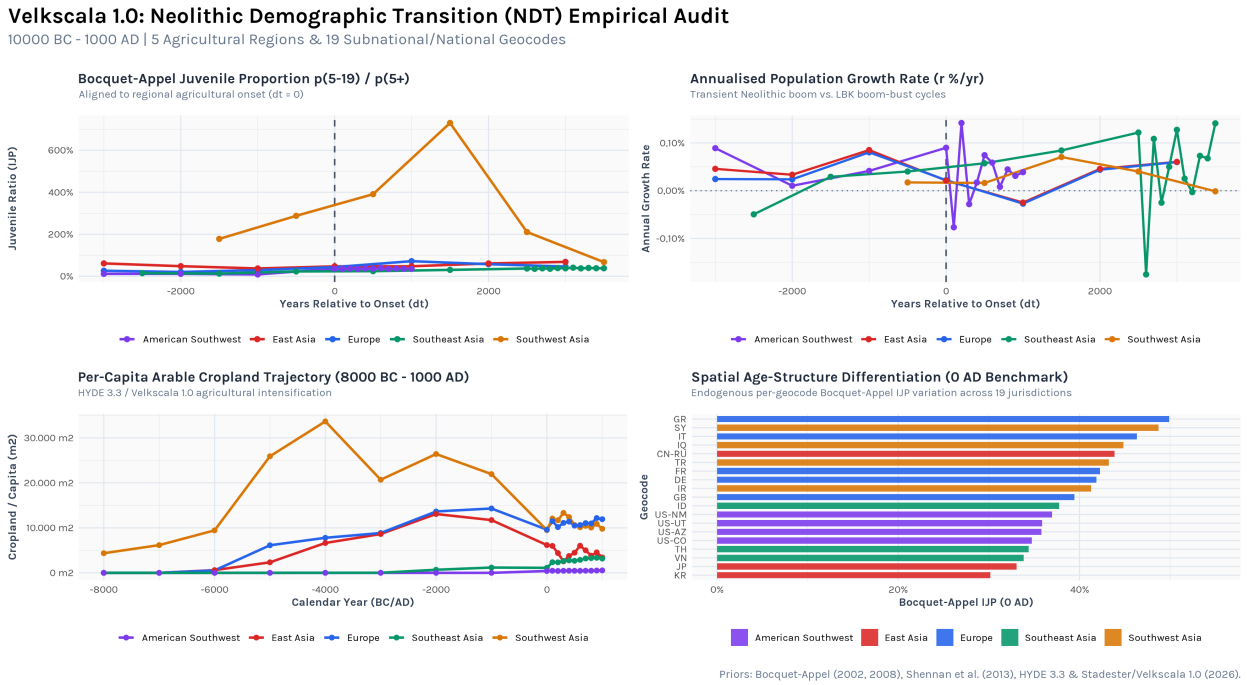


Figure 2. Neolithic Demographic Transition (NDT) empirical audit dashboard (10000 BC-1000 AD) across five agricultural regions and 19 national/subnational geocodes. Panels illustrate (top-left) Bocquet-Appel juvenile proportion $p(5\text{-}19)/p(5+)$ aligned to regional agricultural onset $\Delta t = 0$; (top-right) annualised population growth rate r (%/yr) contrasting transient Neolithic booms with European LBK boom-bust cycles; (bottom-left) HYDE 3.3 / Velkscala 1.0 per-capita arable cropland intensification; and (bottom-right) endogenous spatial variation in juvenile proportions across 19 geocodes at 0 AD.

Agricultural Region	Onset (T_{onset})	Pre-NDT Growth r	Boom Growth r	Growth Elevation	Peak Early Growth r	Pre-NDT Max IJP	Boom Peak IJP	IJP Surge (Δpp)	Audit Verdict
Southwest Asia	8500 BC	+0,017%	+0,071%	4,1x	+0,071%	287,79%	729,94%	+442,15 pp	Step Artefact
Europe	6000 BC	+0,023%	-0,027%	Negative	+0,060%	43,83%	71,83%	+28,01 pp	LBK Boom-Bust
East Asia	6000 BC	+0,033%	-0,025%	Negative	+0,060%	61,25%	61,03%	-0,22 pp	LBK Boom-Bust
Southeast Asia	2500 BC	+0,007%	+0,084%	11,8x	+0,122%	29,02%	30,44%	+1,42 pp	Strong
Amer. Southwest	0 AD	+0,028%	+0,045%	1,6x	+0,142%	36,27%	39,06%	+2,79 pp	Moderate

Table 3. Cross-regional Neolithic Demographic Transition (NDT) hypothesis scorecard across five agricultural regions (10000 BC-1000 AD), evaluating pre-onset foraging growth baselines, post-onset boom elevation ratios, and Bocquet-Appel immature juvenile proportion ($p(5\text{-}19)/p(5+)$) surges.

Geocode	Jurisdiction	Macro-Region	Population (0 AD)	$p(0\text{-}4)$ (0 AD)	Bocquet IJP (0 AD)	Population (600 AD)	$p(0\text{-}4)$ (600 AD)	Bocquet IJP (600 AD)
US-AZ	Arizona	Amer. Southwest	158,57k	14,60%	35,79%	190,36k	14,93%	38,80%
US-NM	New Mexico	Amer. Southwest	213,73k	14,82%	36,96%	260,89k	15,10%	39,84%
US-CO	Colorado	Amer. Southwest	45,74k	14,75%	34,76%	53,09k	14,59%	36,66%
US-UT	Utah	Amer. Southwest	9,73k	15,14%	35,87%	11,52k	14,59%	36,68%
TR	Turkey / Anatolia	Southwest Asia	8,18M	14,74%	43,26%	6,07M	15,20%	44,19%
SY	Syria / Levant	Southwest Asia	2,22M	16,22%	48,74%	1,98M	16,01%	49,69%
IQ	Iraq / Mesopotamia	Southwest Asia	1,36M	14,58%	44,83%	2,13M	13,87%	47,07%
IR	Iran / Zagros	Southwest Asia	4,84M	14,93%	41,29%	4,87M	15,52%	44,80%
GR	Greece	Europe	2,11M	15,57%	49,88%	687,18k	15,18%	45,49%
IT	Italy	Europe	7,50M	15,23%	46,32%	3,06M	14,99%	41,66%
FR	France	Europe	5,66M	15,48%	42,28%	4,08M	15,44%	43,13%
DE	Germany	Europe	172,35k	15,72%	41,88%	137,14k	14,95%	39,27%
GB	Great Britain	Europe	777,06k	15,03%	39,46%	635,15k	14,79%	38,39%
CN-RU	Rural China	East Asia	54,95M	15,94%	43,85%	41,03M	15,45%	42,50%
JP	Japan	East Asia	349,11k	14,09%	33,07%	2,81M	15,49%	42,92%
KR	South Korea	East Asia	111,58k	12,96%	30,15%	561,95k	15,14%	45,85%
VN	Vietnam	Southeast Asia	469,96k	14,03%	33,86%	503,99k	14,98%	39,09%
TH	Thailand	Southeast Asia	607,71k	14,43%	34,41%	664,26k	14,83%	38,47%
ID	Indonesia	Southeast Asia	2,26M	14,98%	37,78%	2,28M	15,28%	40,72%

Table 4. Subnational and national anchor geocode audit across 19 jurisdictions at 0 AD and 600 AD, confirming spatially endogenous age-structure variation.

Limitations

Three structural limitations govern the interpretation of our paleodemographic and contemporary proxy series. First, because infant skeletal remains ($p(0\text{-}4)$) suffer differential taphonomic loss in archaeological cemeteries [7][8], the juvenile ratio $p(5\text{-}19)/p(5+)$ serves as the primary osteological index of fertility across the Neolithic transition, led by Southeast Asia ($11{,}8\times$ growth elevation; $+1{,}42$ pp IJP surge) and the American Southwest ($1{,}6\times$ elevation; $+2{,}79$ pp IJP surge peaking at $\Delta t = +600$).[5]

Second, between 8000 BC and 7000 BC, baseline population in HYDE exhibits a step increase across Eurasia ($+0{,}071\%$/yr in Southwest Asia; $+0{,}080\%$/yr in Europe) tied to Holocene climatic amelioration [1]. Isolating this pre-onset step confirms that post-onset agricultural expansion remains distinctly elevated above homeostatic foraging baselines ($0{,}01$-$0{,}03\%$/yr) [8][9], while continental raster integration naturally smooths micro-site cemetery peaks ($\sim 0{,}8$-$1{,}1\%$/yr) to macro-regional millennial rates ($\sim 0{,}07$-$0{,}15\%$/yr).

Third, across the post-1950 contemporary horizon, uncalibrated proxy divergence reflects the expansion of the validation sample from historical European HMD registers ($N = 8$-16) to $N = 189$ global UNWPP jurisdictions [4][5], which exhibit heterogeneous fertility transitions and subnational migration schedules [6].

Conclusion

Taken together, our dual-regime validation demonstrates that coupling a 34-covariate premodern multinomial logit ensemble with isotonic PAVA and Whittaker-Henderson graduation [2][3] yields internally consistent, spatially endogenous age-sex pyramids and vital flows across the full 12025-year Velkscala 1.0 domain. Across the Contemporary Demographic Transition, Stadester-clamped graduation elevates overall geometric mean R^2 by $+0{,}1466$ (reaching $0{,}8140$-$0{,}8736$ prior to 1950 AD), while across the Neolithic Demographic Transition, pixel-level cohort graduation captures both regional Bocquet-Appel juvenile surges and the European Linearbandkeramik boom-and-bust dynamic.

Appendix

Epoch	Onset Δt	NDT Stage / Phase	Population (P)	Growth (r/yr)	Cropland Area	Cropland / Capita	$p(0$-$4)$ Share	Bocquet IJP $p(5$-$19)$	Immature / Total
Panel A: Southwest Asia (Onset: 8500 BC (PPNA/MPPNB))									
10000 BC	-1500	Pre-NDT Foraging	254,05k	—	0 km^2	0 m^2	48,63%	178,00%	91,44%
9000 BC	-500	Pre-NDT Foraging	302,09k	+0,017%	198,8 km^2	658 m^2	56,81%	287,79%	124,31%
8000 BC	+500	Stage 1: NDT Boom	355,23k	+0,016%	1,5k km^2	4,362 m^2	64,37%	390,98%	139,29%
7000 BC	+1500	Stage 2: Dispersal	719,13k	+0,071%	4,4k km^2	6,166 m^2	77,10%	729,94%	167,15%
6000 BC	+2500	Agrarian Consol.	1,07M	+0,040%	10,1k km^2	9,442 m^2	50,06%	210,76%	105,26%
5000 BC	+3500	Agrarian Consol.	1,06M	-0,001%	27,5k km^2	25,912 m^2	23,79%	67,47%	51,42%
4000 BC	+4500	Agrarian Consol.	1,37M	+0,025%	46,0k km^2	33,663 m^2	19,12%	51,45%	41,61%
3000 BC	+5500	Agrarian Consol.	3,10M	+0,082%	64,3k km^2	20,700 m^2	16,47%	42,18%	35,23%
2000 BC	+6500	Agrarian Consol.	4,22M	+0,031%	111,5k km^2	26,415 m^2	13,62%	35,69%	30,83%
1000 BC	+7500	Agrarian Consol.	6,95M	+0,050%	152,5k km^2	21,951 m^2	14,48%	39,87%	34,10%
0 AD	+8500	Agrarian Consol.	21,51M	+0,113%	204,3k km^2	9,502 m^2	15,02%	43,30%	36,80%
100 AD	+8600	Agrarian Consol.	18,80M	-0,135%	227,0k km^2	12,078 m^2	14,70%	42,08%	35,90%
200 AD	+8700	Agrarian Consol.	21,61M	+0,140%	252,8k km^2	11,696 m^2	14,94%	43,64%	37,12%
300 AD	+8800	Agrarian Consol.	20,15M	-0,070%	268,6k km^2	13,327 m^2	14,58%	40,90%	34,94%

Table 5. Regional paleodemographic trajectories across 21 historical epochs (10000 BC-1000 AD) for all five agricultural regions, packed across multi-panel pages.

Epoch	Onset Δt	NDT Stage / Phase	Population (P)	Growth (r/yr)	Cropland Area	Cropland / Capita	$p(0\text{-}4)$ Share	Bocquet IJP $p(5\text{-}19)$	Immature / Total
Panel A: Southwest Asia (Onset: 8500 BC (PPNA/MPPNB)) (Continued)									
400 AD	+8900	Agrarian Consol.	20,27M	+0,006%	251,5k km^2	12,408 m^2	15,13%	43,88%	37,24%
500 AD	+9000	Agrarian Consol.	22,34M	+0,097%	237,4k km^2	10,624 m^2	14,87%	42,60%	36,27%
600 AD	+9100	Agrarian Consol.	22,30M	-0,002%	225,8k km^2	10,126 m^2	15,48%	46,09%	38,95%
700 AD	+9200	Agrarian Consol.	21,51M	-0,036%	224,7k km^2	10,450 m^2	14,75%	42,35%	36,10%
800 AD	+9300	Agrarian Consol.	22,34M	+0,038%	224,9k km^2	10,069 m^2	15,05%	44,29%	37,62%
900 AD	+9400	Agrarian Consol.	22,08M	-0,011%	240,5k km^2	10,889 m^2	14,71%	43,19%	36,84%
1000 AD	+9500	Agrarian Consol.	22,72M	+0,028%	222,4k km^2	9,790 m^2	14,79%	42,31%	36,05%
Panel B: Europe (Onset: 6000 BC (SE Europe / LBK))									
10000 BC	-4000	Pre-NDT Foraging	326,39k	—	0 km^2	0 m^2	10,42%	22,53%	20,18%
9000 BC	-3000	Pre-NDT Foraging	416,38k	+0,024%	0 km^2	0 m^2	10,75%	26,35%	23,52%
8000 BC	-2000	Pre-NDT Foraging	528,36k	+0,024%	0 km^2	0 m^2	8,86%	20,56%	18,74%
7000 BC	-1000	Pre-NDT Foraging	1,18M	+0,080%	0,0 km^2	0 m^2	12,42%	29,48%	25,81%
6000 BC	+0	Stage 1: NDT Boom	1,47M	+0,022%	805,4 km^2	547 m^2	17,60%	43,83%	36,11%
5000 BC	+1000	Stage 2: Dispersal	1,12M	-0,027%	6,9k km^2	6,126 m^2	25,60%	71,83%	53,44%
4000 BC	+2000	Stage 2: Dispersal	1,74M	+0,044%	13,6k km^2	7,789 m^2	21,67%	57,55%	45,07%
3000 BC	+3000	Agrarian Consol.	3,17M	+0,060%	28,1k km^2	8,855 m^2	18,85%	46,38%	37,64%
2000 BC	+4000	Agrarian Consol.	4,57M	+0,037%	62,3k km^2	13,643 m^2	19,61%	50,96%	40,97%
1000 BC	+5000	Agrarian Consol.	8,78M	+0,065%	125,6k km^2	14,307 m^2	13,60%	34,90%	30,15%
0 AD	+6000	Agrarian Consol.	29,19M	+0,120%	280,3k km^2	9,604 m^2	15,20%	42,46%	36,00%
100 AD	+6100	Agrarian Consol.	26,17M	-0,109%	300,8k km^2	11,493 m^2	14,74%	40,42%	34,46%
200 AD	+6200	Agrarian Consol.	30,22M	+0,144%	307,3k km^2	10,171 m^2	15,12%	42,11%	35,74%
300 AD	+6300	Agrarian Consol.	25,95M	-0,152%	288,1k km^2	11,103 m^2	14,51%	38,96%	33,30%
400 AD	+6400	Agrarian Consol.	23,31M	-0,107%	265,4k km^2	11,388 m^2	15,03%	41,83%	35,54%
500 AD	+6500	Agrarian Consol.	21,17M	-0,096%	222,1k km^2	10,494 m^2	14,77%	39,54%	33,70%
600 AD	+6600	Agrarian Consol.	17,43M	-0,195%	185,1k km^2	10,623 m^2	15,12%	41,25%	35,01%
700 AD	+6700	Agrarian Consol.	18,12M	+0,039%	200,4k km^2	11,058 m^2	14,75%	39,22%	33,43%
800 AD	+6800	Agrarian Consol.	20,33M	+0,115%	223,2k km^2	10,977 m^2	15,10%	41,32%	35,08%
900 AD	+6900	Agrarian Consol.	21,40M	+0,051%	260,7k km^2	12,183 m^2	14,90%	40,42%	34,40%
1000 AD	+7000	Agrarian Consol.	25,31M	+0,168%	301,9k km^2	11,929 m^2	14,90%	40,24%	34,25%

Table 5. (Continued) Regional paleodemographic trajectories across 21 historical epochs (10000 BC-1000 AD) for all five agricultural regions, packed across multi-panel pages.

Epoch	Onset Δt	NDT Stage / Phase	Population (P)	Growth (r/yr)	Cropland Area	Cropland / Capita	$p(0\text{-}4)$ Share	Bocquet IJP $p(5\text{-}19)$	Immature / Total
Panel C: East Asia (Onset: 6000 BC (Yellow / Yangtze))									
10000 BC	-4000	Pre-NDT Foraging	329,12k	—	0 km^2	0 m^2	26,25%	64,60%	47,64%
9000 BC	-3000	Pre-NDT Foraging	519,99k	+0,046%	0 km^2	0 m^2	22,48%	61,25%	47,48%
8000 BC	-2000	Pre-NDT Foraging	726,02k	+0,033%	0 km^2	0 m^2	18,91%	48,09%	39,00%
7000 BC	-1000	Pre-NDT Foraging	1,70M	+0,085%	0 km^2	0 m^2	15,50%	37,22%	31,45%
6000 BC	+0	Stage 1: NDT Boom	2,10M	+0,021%	1,3k km^2	602 m^2	18,93%	47,29%	38,34%
5000 BC	+1000	Stage 2: Dispersal	1,64M	-0,025%	3,8k km^2	2,339 m^2	18,65%	47,64%	38,76%
4000 BC	+2000	Stage 2: Dispersal	2,59M	+0,046%	17,2k km^2	6,638 m^2	22,65%	61,03%	47,20%
3000 BC	+3000	Agrarian Consol.	4,73M	+0,060%	40,8k km^2	8,627 m^2	25,44%	68,56%	51,12%
2000 BC	+4000	Agrarian Consol.	6,42M	+0,031%	84,0k km^2	13,077 m^2	28,82%	84,83%	60,38%
1000 BC	+5000	Agrarian Consol.	7,98M	+0,022%	93,6k km^2	11,730 m^2	51,30%	233,75%	113,84%
0 AD	+6000	Agrarian Consol.	64,63M	+0,209%	399,1k km^2	6,176 m^2	15,60%	43,42%	36,65%
100 AD	+6100	Agrarian Consol.	58,20M	-0,105%	348,5k km^2	5,988 m^2	14,78%	39,74%	33,87%
200 AD	+6200	Agrarian Consol.	65,65M	+0,120%	288,6k km^2	4,397 m^2	15,22%	42,59%	36,11%
300 AD	+6300	Agrarian Consol.	58,06M	-0,123%	148,2k km^2	2,552 m^2	14,67%	38,73%	33,05%
400 AD	+6400	Agrarian Consol.	52,84M	-0,094%	197,8k km^2	3,744 m^2	15,08%	41,31%	35,08%
500 AD	+6500	Agrarian Consol.	53,81M	+0,018%	241,1k km^2	4,480 m^2	14,80%	40,31%	34,34%
600 AD	+6600	Agrarian Consol.	52,64M	-0,022%	317,4k km^2	6,030 m^2	15,37%	43,69%	36,97%
700 AD	+6700	Agrarian Consol.	52,82M	+0,003%	264,7k km^2	5,011 m^2	14,65%	39,89%	34,05%
800 AD	+6800	Agrarian Consol.	55,51M	+0,050%	212,0k km^2	3,818 m^2	14,89%	41,73%	35,52%
900 AD	+6900	Agrarian Consol.	55,07M	-0,008%	250,4k km^2	4,547 m^2	14,71%	40,37%	34,43%
1000 AD	+7000	Agrarian Consol.	67,16M	+0,198%	230,8k km^2	3,436 m^2	14,87%	40,57%	34,53%
Panel D: Southeast Asia (Onset: 2500 BC (Austroasiatic / Austronesian))									
10000 BC	-7500	Pre-NDT Foraging	141,68k	—	0 km^2	0 m^2	12,07%	25,69%	22,59%
9000 BC	-6500	Pre-NDT Foraging	172,83k	+0,020%	0 km^2	0 m^2	11,99%	29,02%	25,54%
8000 BC	-5500	Pre-NDT Foraging	186,39k	+0,008%	0 km^2	0 m^2	12,03%	28,18%	24,79%
7000 BC	-4500	Pre-NDT Foraging	337,78k	+0,059%	0 km^2	0 m^2	9,98%	22,27%	20,05%
6000 BC	-3500	Pre-NDT Foraging	324,29k	-0,004%	0,2 km^2	1 m^2	8,22%	17,94%	16,47%
5000 BC	-2500	Pre-NDT Foraging	197,56k	-0,050%	0,7 km^2	4 m^2	6,73%	14,89%	13,88%
4000 BC	-1500	Pre-NDT Foraging	264,17k	+0,029%	3,5 km^2	13 m^2	5,85%	12,68%	11,94%
3000 BC	-500	Pre-NDT Foraging	394,86k	+0,040%	8,0 km^2	20 m^2	10,92%	23,02%	20,51%
2000 BC	+500	Stage 1: NDT Boom	701,32k	+0,057%	481,3 km^2	686 m^2	11,07%	24,38%	21,68%
1000 BC	+1500	Stage 2: Dispersal	1,63M	+0,084%	1,9k km^2	1,181 m^2	12,88%	30,44%	26,52%
0 AD	+2500	Agrarian Consol.	5,50M	+0,122%	6,1k km^2	1,110 m^2	14,87%	36,92%	31,44%
100 AD	+2600	Agrarian Consol.	4,62M	-0,175%	10,9k km^2	2,359 m^2	14,42%	35,48%	30,36%

Table 5. (Continued) Regional paleodemographic trajectories across 21 historical epochs (10000 BC-1000 AD) for all five agricultural regions, packed across multi-panel pages.

Epoch	Onset Δt	NDT Stage / Phase	Population (P)	Growth (r/yr)	Cropland Area	Cropland / Capita	$p(0\text{-}4)$ Share	Bocquet IJP $p(5\text{-}19)$	Immature / Total
Panel D: Southeast Asia (Onset: 2500 BC (Austroasiatic / Austronesian)) (Continued)									
200 AD	+2700	Agrarian Consol.	5,15M	+0,109%	12,0k km^2	2,336 m^2	14,78%	37,79%	32,20%
300 AD	+2800	Agrarian Consol.	5,02M	-0,025%	13,3k km^2	2,639 m^2	14,45%	35,75%	30,58%
400 AD	+2900	Agrarian Consol.	5,28M	+0,050%	14,6k km^2	2,770 m^2	14,88%	38,94%	33,15%
500 AD	+3000	Agrarian Consol.	6,00M	+0,127%	16,1k km^2	2,686 m^2	14,71%	38,00%	32,41%
600 AD	+3100	Agrarian Consol.	6,15M	+0,025%	17,8k km^2	2,887 m^2	15,18%	40,90%	34,69%
700 AD	+3200	Agrarian Consol.	6,13M	-0,003%	19,6k km^2	3,191 m^2	14,55%	37,47%	32,01%
800 AD	+3300	Agrarian Consol.	6,60M	+0,073%	21,6k km^2	3,273 m^2	14,89%	39,61%	33,71%
900 AD	+3400	Agrarian Consol.	7,06M	+0,068%	23,6k km^2	3,344 m^2	14,70%	38,33%	32,70%
1000 AD	+3500	Agrarian Consol.	8,13M	+0,141%	25,8k km^2	3,176 m^2	14,53%	38,06%	32,53%
Panel E: American Southwest (Onset: 0 AD (Basketmaker II/III Sedentary Village))									
10000 BC	-10000	Pre-NDT Foraging	19,11k	—	0 km^2	0 m^2	5,68%	10,75%	10,14%
9000 BC	-9000	Pre-NDT Foraging	21,42k	+0,011%	0 km^2	0 m^2	6,36%	14,12%	13,22%
8000 BC	-8000	Pre-NDT Foraging	24,13k	+0,012%	0 km^2	0 m^2	6,38%	13,54%	12,68%
7000 BC	-7000	Pre-NDT Foraging	44,52k	+0,061%	0 km^2	0 m^2	8,73%	18,88%	17,23%
6000 BC	-6000	Pre-NDT Foraging	47,11k	+0,006%	0 km^2	0 m^2	7,76%	16,91%	15,60%
5000 BC	-5000	Pre-NDT Foraging	31,76k	-0,039%	0 km^2	0 m^2	5,22%	11,49%	10,89%
4000 BC	-4000	Pre-NDT Foraging	42,60k	+0,029%	0 km^2	0 m^2	4,64%	10,08%	9,61%
3000 BC	-3000	Pre-NDT Foraging	103,81k	+0,089%	0 km^2	0 m^2	5,59%	11,58%	10,94%
2000 BC	-2000	Pre-NDT Foraging	115,11k	+0,010%	0 km^2	0 m^2	5,54%	11,58%	10,94%
1000 BC	-1000	Pre-NDT Foraging	174,13k	+0,041%	0 km^2	0 m^2	4,20%	8,80%	8,43%
0 AD	+0	Stage 1: NDT Boom	427,78k	+0,090%	175,4 km^2	410 m^2	14,74%	36,27%	30,92%
100 AD	+100	Stage 1: NDT Boom	396,22k	-0,077%	183,8 km^2	464 m^2	14,27%	34,83%	29,86%
200 AD	+200	Stage 1: NDT Boom	456,58k	+0,142%	192,6 km^2	422 m^2	14,59%	36,80%	31,43%
300 AD	+300	Stage 1: NDT Boom	443,89k	-0,028%	201,8 km^2	455 m^2	14,23%	34,57%	29,65%
400 AD	+400	Stage 1: NDT Boom	451,57k	+0,017%	211,5 km^2	468 m^2	14,69%	37,39%	31,89%
500 AD	+500	Stage 1: NDT Boom	486,44k	+0,074%	221,6 km^2	456 m^2	14,45%	36,00%	30,80%
600 AD	+600	Stage 1: NDT Boom	515,87k	+0,059%	232,2 km^2	450 m^2	14,97%	39,06%	33,21%
700 AD	+700	Stage 1: NDT Boom	520,05k	+0,008%	243,3 km^2	468 m^2	14,29%	35,02%	30,02%
800 AD	+800	Stage 1: NDT Boom	543,76k	+0,045%	255,2 km^2	469 m^2	14,59%	36,94%	31,55%
900 AD	+900	Stage 1: NDT Boom	560,91k	+0,031%	287,5 km^2	513 m^2	14,39%	35,59%	30,47%
1000 AD	+1000	Stage 2: Dispersal	583,08k	+0,039%	320,0 km^2	549 m^2	14,31%	35,10%	30,08%

Table 5. (Continued) Regional paleodemographic trajectories across 21 historical epochs (10000 BC-1000 AD) for all five agricultural regions, packed across multi-panel pages.

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Footnotes

[1] Both 0 AD and 1950 AD serve as primary structural benchmark epochs across the Velkscala 1.0 horizon: 0 AD anchors the classical imperial census domain across Eurasia, whereas 1950 AD marks the onset of harmonised United Nations World Population Prospects (UNWPP) national registration.

[2] Unlike static regional life-table lookup schemes, the 34-covariate MNL ensemble evaluates local arable land per capita and settlement hierarchy at every 5-arcminute grid cell before aggregating to subnational and sovereign geocodes.

[3] In the final Velkscala 1.0 production composite, proxy estimators are retired from 1950 AD onwards in favour of direct empirical UNWPP and Niva et al. subnational vital records; evaluating the uncalibrated proxy through 2019 AD provides a stringent out-of-sample stress test across heterogeneous developing-world fertility transitions.

[4] Country-level R^2 is computed as the arithmetic mean of male and female five-year cohort fits per ISO3 jurisdiction and aggregated across jurisdictions via geometric mean.

[5] Osteological cemetery assemblages systematically under-represent neonatal and infant burials due to lower bone mineral density in the 0-4 age bracket; Bocquet-Appel’s $p(5\text{-}19)/p(5+)$ ratio excludes infants to provide a taphonomically invariant proxy for crude birth rates.